Lamia Tasnim Khan

Chattogram, Bangladesh | +880 1730 491737 | lamiaatasnimkhan@gmail.com
Google Scholar | GitHub | LinkedIn

Research Interests

Natural Language Processing; Large Language Models; Multimodal Learning; Vision-Language Models; AI Agents and Memory; Trustworthy and Explainable AI; Medical AI; Low-Resource Languages (Bangla).

Education

Chittagong University of Engineering & Technology (CUET) *2022–2026*
B.Sc. in Computer Science and Engineering CGPA: **3.79/4.00**

Thesis: Multimodal AI for Early Detection of Parkinson's Disease Supervisor: Prof. AHM Ashfak Habib

Research Experience

Elite Research Lab *Jul. 2026–Present*
Research Assistant Bangladesh

- Conduct research across four ongoing projects under the supervision of Md. Kishor Morol: three conference-oriented projects and one journal project.
- Current research spans Bengali multimodal understanding, deceptive Bangla SMS, and memory failures in LLM-based agents.

Natural Language Processing Lab, CUET *2024–Present*
Undergraduate Research Collaborator Bangladesh

- Conduct research in natural language processing, large language models, and multimodal learning.
- Contributed to international shared-task research resulting in publications at LT-EDI 2025 and SemEval-2026.
- Developed and evaluated multimodal and LLM-based approaches for language understanding and multilingual NLP tasks.

Publications

[1] M. A. Chowdhury, **L. T. Khan**, M. S. Hasan, A. Dey. *CUET_320@LT-EDI-2025: A Multimodal Approach for Misogyny Meme Detection in Chinese Social Media.* In *Proceedings of the 5th Conference on Language, Data and Knowledge (LT-EDI Workshop)*, 2025.

[2] F. Fariha, **L. Khan**, M. A. Chowdhury, K. Ahmed, M. M. Hoque. *CUET320 at SemEval-2026 Task 10: Few-Shot Large Language Models for Psycholinguistic Marker Extraction and Conspiracy Detection.* In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

[3] M. A. Chowdhury, **L. Khan**, F. Fariha, S. H. Shohan, M. M. Hoque. *CUETClashing at SemEval-2026 Task 1: Multilingual Joke Generation Under Lexical and Topical Constraints Using Small Instruction-Tuned LLMs.* In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

[4] M. Chowdhury, M. A. Chowdhury, **L. T. Khan**, A. S. S. Kashmary, A. Barua, et al. *Brainchild: An Election and Voter Assistant Smart Mobile Application with Election Fraud Detection, Voter Registration, and Nomination Submission Features.* In *2025 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST)*, 2025.

Manuscripts in Preparation

[1] *Who Said What? A Benchmark for Bengali Comic Conversation Understanding.* Conference manuscript in preparation.

[2] *Human or Machine? Stylometric Analysis and Human Evaluation of AI-Generated Deceptive Bangla SMS.* Conference manuscript in preparation.

[3] *Beyond Observational Attribution: Causal Validation and Identifiability of Memory Failures in LLM Agents.* Conference manuscript in preparation.

Thesis Project

An AI-based Technique for Early Detection of Parkinson's Disease Using Multimodal Data
2025–Present
Python, PyTorch, Transformers, Medical AI

- Investigated multimodal learning for early Parkinson's disease detection using heterogeneous clinical and biomedical data.
- Designed modality-specific deep learning encoders and Transformer-based multimodal fusion strategies.
- Explored SHAP and Grad-CAM for improving model interpretability.

Selected Technical Projects

AgroSmart AI
Python, CNN, Flask, React

- Developed a CNN-based soil classification system and integrated predictions into a crop recommendation application.

Niramoy
Kotlin, Firebase

- Developed an Android healthcare application for nurse booking and patient service management using Firebase authentication and database services.

Research Awards & Achievements

- **1st Place**, LT-EDI@LDK 2025 International Shared Task on Multimodal Misogyny Meme Detection, 2025.
- **Finalist**, HackCSB 2024 — Top 7 teams worldwide among 160+ teams, 2024.
- **Finalist**, IEEE Computer Society CUET Intra Machine Learning Contest (LLM-based Physics QA), 2025.
- **Finalist**, Nodescape Multi-stage Hackathon, 2025.

Academic Service

- **Reviewer**, LT-EDI 2025 Shared Task (Multimodal Natural Language Processing), 2025.
- **Volunteer**, Technical Sessions and Academic Events, 2024–Present.

Technical Skills

Programming: Python, C++, JavaScript, SQL, Kotlin
AI/ML: Machine Learning, Deep Learning, NLP, Multimodal Learning, Large Language Models, Generative AI, Explainable AI
Frameworks/Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Flask, Node.js, Express.js

Data/Infrastructure: MongoDB, Pinecone, Git, GitHub, Docker
Research/Documentation: LaTeX, Figma

Leadership & Activities

- **Vice President**, CUET Career Club.
- **SheSTEM Mentee**.

Additional Experience

The Tork *Nov. 2024–Mar. 2025*
Intern Bangladesh

- Developed multimedia and branding content using AI-assisted tools and Figma.
- Collaborated with marketing and development teams and coordinated project workflows using Jira and Slack.

Selected Training

- Supervised Machine Learning: Regression and Classification, Coursera.
- Python for Data Science, Kaggle Learn.
- EDGE Mobile Application Development Program, ICT Division, Bangladesh.

References

Prof. AHM Ashfak Habib
Professor, Department of Computer Science & Engineering
Chittagong University of Engineering & Technology (CUET)
Thesis Supervisor
ashfak@cuet.ac.bd

Dr. Mohammed Moshiul Hoque
Professor, Department of Computer Science & Engineering
Director, CUET IT Business Incubator
Chittagong University of Engineering & Technology (CUET)
moshiul_240@cuet.ac.bd

Md. Kishor Morol
Chief Executive Officer, Elite Research Lab
AI Safety Researcher, Meta
Incoming Researcher, Anthropic
Research Supervisor
kishor@elitelab.ai

Declaration

I acknowledge that the above information is true and correct.

Sincerely,



Lamia Tasnim Khan